

AI WeatherWise API

Project title: “ AI WeatherWise Api ”

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Team Members:

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4: Project Planning Phase

Step 1 – Requirement Analysis

- Identify the functional and technical requirements of the AI WeatherWise API.
- Define features such as user registration, login, weather information, favorite locations, and AI-based recommendations.
- Identify the required hardware, software, APIs, and database.

Step 2 – Technology Selection

- Select **Node.js** and **Express.js** for backend development.
- Select **MongoDB** and **Mongoose** for database management.
- Use **JWT** and **bcryptjs** for authentication and password security.
- Use **Google Gemini AI** for weather summaries and recommendations.
- Use **Postman** for API testing.

Step 3 – Project Setup

- Create the project folder and initialize the Node.js project.
- Install the required packages and dependencies.
- Configure environment variables for database and API credentials.
- Set up the Express.js server.

Step 4 – Database Planning

- Create the MongoDB database.
- Design the **User** and **Location/Favorite Location** collections.
- Establish the required fields and relationships.
- Configure Mongoose for database connectivity.

Step 5 – Backend Development

- Develop RESTful API routes.
- Implement user registration and login.
- Add JWT-based authentication and authorization.
- Develop weather and favorite-location APIs.

Step 6 – AI Integration

- Integrate Google Gemini AI with the backend.
- Process weather information and user queries.
- Generate weather summaries, suggestions, and recommendations.

Step 7 – Security Implementation

- Hash passwords using bcryptjs.
- Protect APIs using JWT authentication.
- Validate and sanitize user inputs.
- Store sensitive credentials using environment variables.
- Implement centralized error handling.

Step 8 – API Testing

- Test registration and login APIs using Postman.
- Test weather and favorite-location APIs.
- Check valid and invalid inputs.
- Verify authentication, authorization, and error responses.

Step 9 – Integration and Debugging

- Connect the frontend/API clients with the backend.
- Verify MongoDB, weather API, and Gemini AI integration.
- Identify and fix errors.
- Test the complete system workflow.

Step 10 – Documentation and Finalization

- Document the API endpoints and project workflow.
- Prepare screenshots and testing results.
- Organize project files phase-wise.
- Upload the completed project and documents to the public GitHub repository.